

Bridging the Integrity Gap: Towards AI-assisted Design Research

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ABSTRACT

Design research involves the human-centered analysis of substantial volumes of qualitative data, presenting a promising application for emerging generative AI technologies. While AI-assisted qualitative research remains an extensively explored topic within the Human-Computer Interaction (HCI) discourse, the integration of AI into design research, differentiated by its intrinsic reliance on empathy and intuition, remains markedly underexplored. This paper reports on a qualitative study with 13 designers from a range of design disciplines, investigating the processes by which insights are cultivated in design research and the extent to which these processes may be effectively augmented or potentially undermined by AI interventions. The findings suggest that design research relies heavily on the tacit knowledge of designers, underlining the importance of designer-AI alignment. Drawing upon these findings and existing literature on human-AI trust, this paper lays the foundation for further inquiry into AI-assisted design research, by (1) identifying the development of intrinsic trust through designer-AI alignment as a central objective; and (2) introducing the conceptual framework of the Integrity Gap to motivate further studies.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; *HCI theory, concepts and models.*

KEYWORDS

designer-AI alignment, design research, trust

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1 INTRODUCTION

In recent decades, the growing complexity of design briefs has prompted a transition across a variety of design disciplines towards evidence-based approaches, where the creation of design outcomes, or *artifacts*, is informed by systematic research methods [18]. This paper refers to *design research* as the process of predominately qualitative sense-making that designers undertake in design projects, also referred to as "research for design" by some scholars [22]. The

aim of this paper is to unearth and explore questions surrounding the imminent and ongoing integration of AI into this process.

Design seeks answers to complex challenges often characterised by open-endedness, where there exists no singular definitive solution [4]. Consequently, the work of a designer involves a high degree of uncertainty [13]. To navigate this uncertainty, design research adopts iterative methodologies encompassing qualitative exploration and synthesis, paralleling approaches found in academic qualitative research.

Nevertheless, the processes for cultivating insights in design research differ significantly from those employed in other forms of qualitative research. Krippendorff [12] introduces the concept of the *semantic turn*, which creates a separation between artifacts and the meanings they carry for humans, positioning designers as "experts in the meanings of artifacts". Thus, he proposes that design research operates on a semantic level, aiming to develop an understanding of other humans' understandings. Consequently, insights are interpretations that cannot easily be validated through their correspondence with observations or other data, as they are inadvertently influenced by the designer's "intuitive understanding of phenomena" [21]. Furthermore, where academic qualitative research often aims to establish an objective view into how things *are*, design research is more concerned with investigating potential ideas for how things *could be* [5, 13]; design is fundamentally a pragmatic field focused on "making changes happen" [12]. As a result, the *insights* sought by design research are distinct from the *truths* sought by academic research. Therefore, while there exists plenty of literature on AI-assisted qualitative research in an academic context [6–9, 11, 16, 23, 24], the impact of AI assistance warrants a dedicated exploration in the context of design research, with its unique practises and objectives.

Given the practical orientation of design as a discipline, this paper appropriately investigates AI-assisted design research through an empirical case study of designers' approaches to interviews and their subsequent analysis. Semi-structured interviews are a widely embraced method in design research, used for collecting valuable data about people's perspectives, preferences, and experiences—all of which constitute pivotal elements informing the design process. The nature of these interviews leads to the generation of extensive volumes of unstructured and diverse qualitative data, which designers make sense of using *design tools*, such as the affinity diagram [15]. These tools, which are often manual in nature, facilitate information synthesis by scaffolding the analytical process, and thus help designers navigate the uncertainties inherent to their practice.

Recent advances in language models have significantly expedited research efforts in the domain of automatic summarisation, extending its capabilities to challenging conversational contexts [20] and in some tasks to human levels of performance [25]. Some of these technologies have already been adopted by commercial interview

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tools used by designers, such as Marvin¹ and DoveTail², promising to deliver automatically generated insights from interview recordings. Nevertheless, the extent to which highly automated AI tools are compatible with the domain of design research remains an open question, owing to the unique nature of design insights and the uncertainties inherent to design practice. The challenge lies in the difficulty of evaluating the correctness of insights, which underlines the importance of designers' trust towards them. Consequently, there is a pressing need to investigate the potential impact of automatic synthesis on this trust, as it holds significant implications for the effectiveness and credibility of AI-assisted design research.

2 METHOD

Table 1: Participant demographics. *Combined number of years spent studying or working in design. **BD: Business Design; GD: Graphic Design, ID: Industrial Design; SD: Service Design; UX: UX Design.

Label	Design experience*	Design disciplines**	Context. inquiry	Interview
P1	4-9 years	SD, UX	✓	✓
P2	10+ years	BD, GD, ID, SD	✓	✓
P3	10+ years	ID, SD		✓
P4	10+ years	BD, GD, SD, UX		✓
P5	1-3 years	BD, SD		✓
P6	10+ years	ID, SD		✓
P7	4-9 years	SD		✓
P8	10+ years	ID, SD		✓
P9	10+ years	GD, ID, SD		✓
P10	1-3 years	BD, SD		✓
P11	1-3 years	SD		✓
P12	4-9 years	GD, SD		✓
P13	10+ years	SD, UX	✓	

This paper reports on a qualitative study involving two rounds of data collection with 13 designers (see Table 1). In the first round, three participants took part in contextual inquiry sessions where their interviewing and interview analysis practices were observed alongside a service design project. In the second round, 12 participants took part in semi-structured interviews. The participants had various levels of expertise in a wide range of design disciplines. Ten participants were recruited from the employees of a design agency and three from a pool of graduate-level design students. The participants' ages ranged from 26 to 46 with a median age of 30. Among the participants, ten were from Finland, with one participant each from Germany, India, and the United Kingdom. The sample included nine women and four men.

The first round of contextual inquiry sessions was conducted to gain an overall understanding of design research as a complex process consisting of multiple interconnected phases, often spread across several weeks. The designers were observed over five weeks

as they conducted 12 semi-structured interviews and several data analysis sessions throughout the duration of a service design project. In addition to better understanding the context in which design research operates, the intended outcome was to identify existing behaviours and opportunities for AI-assistance that could be studied in depth in the subsequent interviews.

The semi-structured interviews delved deeper into insights and themes identified in the first round of inquiry. Each hour-long interview was structured into two halves: the first focused on exploring designers' practices and experiences across various stages of conducting and analyzing interviews, aiming to enhance understanding of how insights are generated. In the latter half, participants were presented with vignettes [1] of hypothetical AI-assisted interview scenarios to provoke discussion on specific types of AI support.

The interviews, along with notes from the observation, were transcribed and pseudonymised. Subsequently, the author conducted a reflexive thematic analysis [2] on the dataset. The resulting themes are discussed in the following section.

3 FINDINGS

3.1 Levels of analysis

A significant portion of the design projects discussed by participants entailed conducting multiple interviews across a range of individuals, yielding a broad spectrum of perspectives for subsequent data analysis. Based on the practices described by participants, three distinct levels of analysis emerged: *contextual*, *comparative*, and *holistic*. These levels differ in the breadth and depth of data analysed, the aims of the analysis, and the particular analytical methods employed by the designers.

3.1.1 Contextual analysis. This paper refers to contextual analysis as the practice of analysing individual interviews while maintaining a deep understanding of the context in which the interview took place. Many participants highlighted the importance of retaining access to full chronological transcriptions or extensive notes. For example, P1 had a practice of beginning any interview analysis by reading through the notes for each interview: "With quantitative data, it's easier to compare individual questions across different people, but with qualitative data, it is more important to analyse responses as a whole." This idea of completeness was reflected in other interviews, whose participants preferred to "write everything down" (P4) or "have this transcription of almost each and every word the person has said." (P6)

For P1 and many other participants, being able to ground the interviewee's words in their full context was crucial to "understanding the interviewee as a person". Fostering empathy for the end user is a common goal in design [13], to understand how their needs can be met through the creation of new artifacts. P6 distinguished a notion of "contextual understanding" from the practical insights that directly address the project's research questions: "I think there are the key insights for solving a problem in the design process, or there are some key insights for just kind of getting your head into the context that sometimes can be a completely different world than what you yourself as a human live in... so there's that extra layer of data that you're gathering as an interviewer in addition to the actual data that's in words on a transcription." This additional

¹<https://heymarvin.com/>

²<https://dovetail.com/>

layer of tacit knowledge was rarely formalised, but rather viewed as something deeply personal to the designer, and complementary to the factual information that is stored explicitly as transcriptions or notes. P4 referred to this as "thick data": information that "connects with you, with your own emotions".

3.1.2 Comparative analysis. While contextual analysis focuses on empathising with a single interviewee, comparative analysis seeks to identify similarities and differences across views expressed by multiple interviewees. Often, the purpose of comparative analysis was to "validate what [another] person said" (P5), in order to corroborate early insights identified in contextual analysis. So while designers aimed to empathise with interviewees as deeply as possible on an individual level, there was a parallel effort to discover overarching insights that apply to a number of different people. Aside from validation, another purpose of comparative analysis was to identify the most significant insights to maximise impact: for P4, interview analysis was about identifying "what many people might be saying in different ways" in order to identify a "core concept" instead of incremental improvements.

3.1.3 Holistic analysis. Both contextual and comparative analysis provide important insights that inform the design process with depth and breadth, respectively. Holistic analysis is the idealised synthesis of both analytical processes, where the designer holds a simultaneously deep and broad understanding of the design space, understanding the interviewees' individual perspectives and the connections between them, as well as the wider business and societal setting of the project. While many participants strived to obtain such a "holistic picture" (P6) of the design space, they often found it difficult due to a lack of resources. This was particularly the case in the common scenario where interviews and their analysis were fragmented across multiple sessions with different designers. P3 had often worked in such scenarios, and noted a "risk of losing some of the qualitative understanding from the people who were actually there during the interviews", advocating for at least one designer who takes part in all interviews in a project and understands the "big picture". Many more participants alluded to the importance of agency and ownership, such as P10, who found themselves distrusting interview notes when they had not participated in the interviews: "The [note-taker] had chosen what they think is important, so my insights from the notes were limited to those notes and I couldn't see for myself."

3.2 Tacit knowledge guides synthesis

It is evident from the preceding discussion that design research involves the synthesis of information in various degrees and forms, and from multiple perspectives. The primary motivation for this synthesis is the sheer amount of qualitative data that is collected in the design research process, which many participants felt overwhelmed by at times—P8 referred to this as a state of "analysis paralysis". Given the exploratory nature of the semi-structured interview format, participants noted a high variance in the informational value of the collected data: "It could be that like 80% is quite useless generic stuff, but then you get like 20% where you get something new, insightful" (P6). Synthesis aims to distill this information into its most valuable parts: "There's a lot of interesting

stuff that you discover; but how do you simplify that and how do you report the *key insights*, the most important things?" (P2)

Given the central role of synthesis in design research, it is essential that any automatic tool effectively discerns insightful content in alignment with designer preferences. Yet, the prioritisation of insights necessitates a method for their identification and evaluation, which many participants felt was an ability largely acquired through experience and heavily reliant on intuition. For instance, P3 and P6, each with over a decade of industry experience, habitually highlighted the most intuitively interesting notes during the interview note-taking process, demonstrating high proficiency in spontaneous prioritisation. In contrast, P5, with considerably less experience, found the task more challenging: "Taking notes absolutely can be tricky... it can absolutely be hard to find what are the key points? What are the main insights here?"

Much of this intuitive skill is grounded in tacit knowledge; as much as insights are shaped by the words of the interviewee, they are also shaped by the context of the design project, as well as the designer's intuition. P2 listed many other factors that contribute to insights in addition to the interviewee's statements: "they're usually a combination of what you learned previously or what do you know, what do you maybe assume, what are the targets, the KPIs, and then what the customers tell." For many participants, personal reflection was another source informing their insights. Multiple participants remarked that they spend a lot of time reflecting on interviews, sometimes even outside of work hours. By allowing their thoughts to "marinate" (P1), the participants were able to add depth to what was said in the interview: "...you need to bring meaning into [the data], and this meaning comes from this reflective process you have; that it's not just quotes." (P4) The participants rarely attempted to formalise this reflection, but it played an important role in shaping their understanding of the design space.

Given that insights are built upon layers of tacit knowledge residing solely in the designer's mind, many participants expressed concerns about the unreliability of their memory. P12 acknowledged that working with their intuition often lead to choosing the "points that stick in [their] mind", a practice they deemed highly vulnerable to personal bias and assumptions. Numerous participants recounted instances where their recollection had benefited from revisiting their notes post-interview. One important aim of this practice, as P1 noted, was to "make sure that interpretations remain grounded in the data", highlighting the necessity of aligning their personal reflective analysis with what the data was saying.

3.3 Tacit knowledge and AI alignment

Many of the participants had some experience with using AI tools in their design practice. While these participants found AI tools useful with menial data analysis tasks such as search and pattern detection, they were hesitant about the idea of using the tools to generate the insights themselves. P2 describes their experience working with a tool that assists with insight generation from large quantities of survey responses: "The machine, by my experience, doesn't give you the insight as such... [You need to] combine the qualitative findings that you discovered already in the interview phase to bring character and more human flavor to the answers: what is the context of this, these findings that the machine kind of

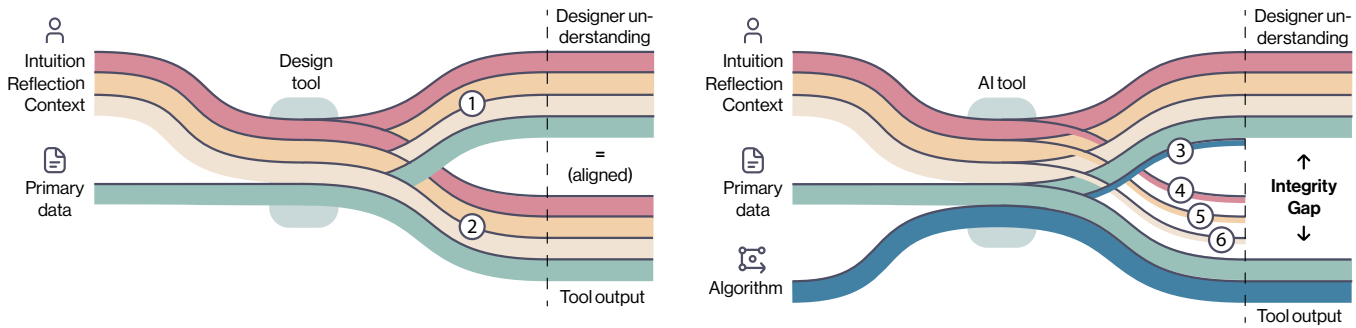


Figure 1: Potential differences in how traditional design tools and AI tools synthesise information, generate insights, and foster a (mis)aligned understanding of the design space between the designer and the tool's output.

helped me to discover?" The same sentiment was echoed by P4 in describing their prior experiments with utilising ChatGPT³ for the construction of affinity diagrams, comparing it to "a very intelligent and analytical person with very little understanding of human life in the room."

Much of this scepticism concerned the lack of contextual understanding of such tools, given that they had no access to the designer's tacit knowledge. Another concern had to do with agency: some participants felt that with AI-generated insights, they could not trust or justify the insights as well as if they themselves had generated them, given that they were "losing that process in the hands of a machine." (P2) While participants found it important that they could impose their tacit knowledge on the analysis, it was equally important for them to use the analysis as an opportunity to *develop* their own understanding: "We can automate the manual work, but we cannot automate the learning, and if I don't learn the insights, then I won't be able to use them for the rest of the project... otherwise it's the AI who has the insights, not me." (P5)

4 DISCUSSION AND FUTURE WORK

This case study reveals a significant challenge for AI-assisted design research, specifically concerning the alignment of analytical processes between designers and AI systems. A key tension in design research takes place between its creative utilisation of the designer's tacit knowledge and its simultaneous aim to achieve evidence-based validity. Consequently, one plausible function of AI could involve aiding designers in aligning their reflective analysis and tacit knowledge with the raw interview data. Conversely, a critical hurdle for the developers of such tools lies in aligning the reasoning and objectives of AI with those of the designer, whose analysis is influenced by numerous factors extending beyond the scope of readily available interview data.

4.1 Human-AI trust in design research

Alignment is closely related to the concept of trust, which has been shown to be a precursor for the successful adoption of automation [14]. Building upon the ABI (Ability, Benevolence, Integrity) framework for interpersonal trust [17], Jacovi et al. [10] formalise the mechanisms that facilitate human-AI trust into notions of *extrinsic*

and *intrinsic* trust. Extrinsic trust is related to ability, as it is built through the evaluation of the AI's capability to accomplish a given task, for example through test sets and performance metrics. Intrinsic trust, on the other hand, is related to integrity, as it results from the user's understanding that the AI's reasoning process aligns with "the user's priors of agreeable reasoning" [10].

Building extrinsic trust requires evaluating the correctness of the AI's output, which is difficult in design research due to its open-ended nature. Indeed, Jiang et al. [11] propose that the "intentionally ambiguous" objective of qualitative research makes it a "unique case for human-AI collaboration" and an unexplored challenge for human-AI trust. Therefore, I propose that **intrinsic trust is a necessity for successful designer-AI collaboration, underlining the importance of alignment and integrity between the designer and the AI.**

4.2 The Integrity Gap

Figure 1 illustrates how traditional design tools synthesise information to produce an aligned understanding of the design space, while AI tools may fail to achieve the same goal, resulting in an *integrity gap* between how the designer perceives the design space and how it is represented in the tool's output.

4.2.1 Traditional design tools facilitate aligned understanding. Designers make use of tools such as the affinity diagram to consolidate an otherwise overwhelming amount of data that is collected during design research. These tools:

- (1) actively involve the designer, developing the designer's understanding of the design space;
- (2) allow the designer to integrate their tacit knowledge into the tool's output, by complementing any primary data, such as interview notes, with their intuition, reflection, and contextual understanding.

4.2.2 AI tools may promote misalignment. Tools building upon advanced AI technologies, such as language models, are a promising addition to the designer's toolkit, as they can help designers analyse large quantities of qualitative data in increasingly sophisticated ways. However, many AI tools strive for heavy automation, in which case there is a risk that neither (1) nor (2) above hold, potentially leading to misalignment, as illustrated by the integrity gap, and ultimately, mistrust towards the AI tool. In order to bridge the

³<https://chat.openai.com/>

integrity gap, I propose the following avenues for future work on AI-assisted design research:

- (3) exposing the tool's reasoning to the designer, for example through explainable AI methods, enabling the designer to learn from its analysis.
- (4) actively involving the designer's intuition in the analysis in order to foster a feeling of agency and ownership over the output.
- (5) supporting and integrating the designer's personal reflection, for example through conversational interfaces.
- (6) supplementing the AI's analysis with contextual information, for example by integrating multi-modal AI [3] to better understand the interview context, or information retrieval [19] to retrieve the context from unstructured data sources.

5 CONCLUSION

Emerging AI technologies, such as language models, can support the goal of design research in achieving a holistic understanding of the design space by synthesising and identifying connections in large quantities of qualitative data. However, such tools must collaborate and share agency with the designer, who is in a unique position to address the integral task of contextual analysis, a practice deeply rooted in the human factors of empathy and personal reflection. Ensuring aligned information synthesis between the designer and the AI contributes to trust towards the AI, which is imperative for its successful integration into design research practice. This paper aims to motivate further research into the topic by proposing the integrity gap as a framework towards designer-AI alignment.

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REFERENCES

- [1] Christiane Atzmüller and Peter M Steiner. 2010. Experimental vignette studies in survey research. *Methodology: European Journal of Research Methods for the Behavioral and Social Sciences* 6, 3 (2010), 128–138. <https://doi.org/10.1027/1614-2241/a000014>
- [2] Virginia Braun and Victoria Clarke. 2006. Using thematic analysis in psychology. *Qualitative Research in Psychology* 3 (2006), 77–101. <https://doi.org/10.1191/1478088706qp0630a>
- [3] Sébastien Bubeck, Varun Chandrasekaran, Ronen Eldan, Johannes Gehrke, Eric Horvitz, Ece Kamar, Peter Lee, Yin Tat Lee, Yuanzhi Li, Scott Lundberg, Harsha Nari, Hamid Palangi, Marco Tulio Ribeiro, and Yi Zhang. 2023. Sparks of Artificial General Intelligence: Early experiments with GPT-4. [arXiv:2303.12712 \[cs.CL\]](https://arxiv.org/abs/2303.12712)
- [4] Kees Dorst. 2011. The core of 'design thinking' and its application. *Design Studies* 32, 6 (2011), 521–532. <https://doi.org/10.1016/j.destud.2011.07.006>
- [5] Melanie Feinberg. 2012. Information Studies, the Humanities, and Design Research: Interdisciplinary Opportunities. In *Proceedings of the 2012 IConference* (Toronto, Ontario, Canada) (iConference '12). Association for Computing Machinery, New York, NY, USA, 18–24. <https://doi.org/10.1145/2132176.2132179>
- [6] Jessica L. Feuston and Jed R. Brubaker. 2021. Putting Tools in Their Place: The Role of Time and Perspective in Human-AI Collaboration for Qualitative Analysis. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW2, Article 469 (2021), 25 pages. <https://doi.org/10.1145/3479856>
- [7] Jie Gao, Yuchen Guo, Toby Jia-Jun Li, and Simon Tangi Perrault. 2023. Col-labCoder: A GPT-Powered Workflow for Collaborative Qualitative Analysis. In *Companion Publication of the 2023 Conference on Computer Supported Cooperative Work and Social Computing* (Minneapolis, MN, USA) (CSCW '23 Companion). Association for Computing Machinery, New York, NY, USA, 354–357. <https://doi.org/10.1145/3584931.3607500>
- [8] Simret Araya Gebreegziabher, Zheng Zhang, Xiaohang Tang, Yihao Meng, Elena L. Glassman, and Toby Jia-Jun Li. 2023. PaTAT: Human-AI Collaborative Qualitative Coding with Explainable Interactive Rule Synthesis. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 362, 19 pages. <https://doi.org/10.1145/3544548.3581352>
- [9] Matt-Heun Hong, Lauren A. Marsh, Jessica L. Feuston, Janet Ruppert, Jed R. Brubaker, and Danielle Albers Szafrir. 2022. Scholastic: Graphical Human-AI Collaboration for Inductive and Interpretive Text Analysis. In *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology* (Bend, OR, USA) (UIST '22). Association for Computing Machinery, New York, NY, USA, Article 30, 12 pages. <https://doi.org/10.1145/3526113.3545681>
- [10] Alon Jacovi, Ana Marasović, Tim Miller, and Yoav Goldberg. 2021. Formalizing Trust in Artificial Intelligence: Prerequisites, Causes and Goals of Human Trust in AI. In *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency* (Virtual Event, Canada) (FAccT '21). Association for Computing Machinery, New York, NY, USA, 624–635. <https://doi.org/10.1145/3442188.3445923>
- [11] Jialun Aaron Jiang, Kandrea Wade, Casey Fiesler, and Jed R. Brubaker. 2021. Supporting Serendipity: Opportunities and Challenges for Human-AI Collaboration in Qualitative Analysis. *Proc. ACM Hum.-Comput. Interact.* 5, CSCW1, Article 94 (2021), 23 pages. <https://doi.org/10.1145/3449168>
- [12] Klaus Krippendorff. 2005. *The Semantic Turn: A New Foundation for Design* (1. ed.). CRC Press, Boca Raton, FL, USA. <https://doi.org/10.4324/9780203299951>
- [13] Klaus Krippendorff. 2007. Design Research, an Oxymoron? In *Design Research Now*, Ralf Michel (Ed.). Birkhäuser, Berlin, Boston, 67–80. https://doi.org/10.1007/978-3-7643-8472-2_5
- [14] John D. Lee and Katrina A. See. 2004. Trust in Automation: Designing for Appropriate Reliance. *Human Factors* 46, 1 (2004), 50–80. <https://doi.org/10.1518/hfes.46.1.50.30392>
- [15] Andrés Lucero. 2015. Using Affinity Diagrams to Evaluate Interactive Prototypes. In *Human-Computer Interaction – INTERACT 2015*, Julio Abascal, Simone Barbosa, Mirko Fetter, Tom Gross, Philippe Palanque, and Marco Winckler (Eds.). Springer International Publishing, Cham, 231–248.
- [16] Megh Marathe and Kentaro Toyama. 2018. Semi-Automated Coding for Qualitative Research: A User-Centered Inquiry and Initial Prototypes. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems* (Montreal QC, Canada) (CHI '18). Association for Computing Machinery, New York, NY, USA, 1–12. <https://doi.org/10.1145/3173574.3173922>
- [17] Roger C. Mayer, James H. Davis, and F. David Schoorman. 1995. An Integrative Model of Organizational Trust. *The Academy of Management Review* 20, 3 (1995), 709–734. <https://doi.org/10.2307/258792>
- [18] Gjoko Muratovski, Ken Friedman, Don Norman, and Steven Heller. 2022. *Research for Designers: A Guide to Methods and Practice* (2. ed.). SAGE Publications, London.
- [19] Marc Najork. 2023. Generative Information Retrieval. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval* (Taipei, Taiwan) (SIGIR '23). Association for Computing Machinery, New York, NY, USA, 1. <https://doi.org/10.1145/3539618.3591871>
- [20] Virgile Rennard, Guokan Shang, Julie Hunter, and Michalis Vazirgiannis. 2023. Abstractive Meeting Summarization: A Survey. *Transactions of the Association for Computational Linguistics* 11 (2023), 861–884. https://doi.org/10.1162/tacl_a_00578
- [21] Cal Swann. 2002. Action Research and the Practice of Design. *Design Issues* 18, 1 (2002), 49–61. <http://www.jstor.org/stable/1512029>
- [22] Katja Tschimmel Violeta Clemente and Fátima Pombo. 2017. A Future Scenario for a Methodological Approach applied to PhD Design Research. Development of an Analytical Canvas. *The Design Journal* 20, sup1 (2017), S792–S802. <https://doi.org/10.1080/14606925.2017.1353025>
- [23] Gregor Wiedemann. 2013. Opening up to Big Data: Computer-Assisted Analysis of Textual Data in Social Sciences. *Historical Social Research / Historische Sozialforschung* 38, 4 (146) (2013), 332–357. <http://www.jstor.org/stable/24142701>
- [24] Ziang Xiao, Xingdi Yuan, Q. Vera Liao, Rania Abdelghani, and Pierre-Yves Oudeyer. 2023. Supporting Qualitative Analysis with Large Language Models: Combining Codebook with GPT-3 for Deductive Coding. In *Companion Proceedings of the 28th International Conference on Intelligent User Interfaces* (Sydney, NSW, Australia) (IUI '23 Companion). Association for Computing Machinery, New York, NY, USA, 75–78. <https://doi.org/10.1145/3581754.3584136>
- [25] Tianyi Zhang, Faisal Ladhak, Esin Durmus, Percy Liang, Kathleen McKeown, and Tatsunori B. Hashimoto. 2024. Benchmarking Large Language Models for News Summarization. *Transactions of the Association for Computational Linguistics* 12 (2024), 39–57. https://doi.org/10.1162/tacl_a_00632